BRIGHTLEARN DATA ANALYTICS

SQL FUNDAMENTALS EXERCISE 4

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Course : BRIGHTLEARN BASIC SQL

SQL UNION AND UNION ALL

01. Combine the account holders from ~~fro~~ both branches into one unique list. Each person should appear only once, even if they hold account at both branches.

```
SELECT account_id, account_holder, city
FROM sandton
UNION
SELECT account_id, account_holder, city
FROM rosebank);
```

account_id	account_holder	city
A001	Nonvula Dlamini	Johannesburg
A002	David Mokoena	Johannesburg
A003	Lerato Sithole	Johannesburg
A004	Peter Nkosi	Johannesburg
A005	Zanele Khumalo	Johannesburg
A006	Thabo Motha	Johannesburg

02. NexBank offers two products ~~pro~~ lists - one for savings accounts and one for current account. Build a unique product cataloque. that shows every product the bank offers, without repeating any product that appears in both lists.

(2)

```
SELECT product_code, product_name,
       'Savings' AS product_type
FROM savings_products
UNION
SELECT product_code, product_name,
       'Current' AS product_type
FROM current_products );
```

Product_code	Product-name	product-type
SV01	Basic Savings	Savings
SV02	Premium Savings	Savings
SV03	Youth Savings	Savings
SV04	Business Savings	Saving
CR01	Standard Current	Current
CR02	Gold Current	Current
CR04	Business Current	Current

Question 03 (FOR CODE SEE UNDER/BELOW Q4) →

The HR team needs a unique list of all staff who work in either Retail Banking or Corporate Banking. Some staff are assigned to both. Each staff member should appear only once.

Staff_id	staff_name	email
S001	Mpho Radebe	mpho01.radebe@nexbank.co.za
S002	Brian Tshabalala	brian.t23@nexbank.co.za
S003	Aisha Patel	aisha.p@nexbank.co.za
S004	Kabelo Moabelo	kabelo.m@nexbank.co.za
S005	Nandi Dube	nandi.d@nexbank.co.za
S006	Sipho Khumalo	sipho.k@nexbank.co.za

Question 4

Marketing needs a unique list of all cities where NexBank serves customers - either through its mobile branches or its digital branch (which serves customers online regardless of city). No city should appear twice.

city_code	city_name	region
C01	Johannesburg	Gauteng
C02	Pretoria	Gauteng
C03	Cape Town	Western Cape
C04	Durban	Kwa-Zulu Natal
C05	Polokwane	Limpopo
C06	Port Elizabeth	Eastern Cape

QUESTION 03

```
(SELECT staff_id, staff_name, email
FROM retail_banking_staff
UNION
SELECT staff_id, staff_name, email
FROM corporate_banking_staff );
```

QUESTION 04

```
(SELECT city_code, city_name, region
FROM mobile_branch_cities
UNION
SELECT city_code, city_name, region
FROM digital_branch_cities );
```

Questions

The digital marketing team ran two campaigns last month - one via push notification and one via in-app banner. Build a unique list of every customer who was targetted. If a customer received both, the must appear only once.

```
(SELECT customer_id, customer_name, segment
FROM push_notifications_targets
UNION
SELECT customer_id, customer_name, segment
FROM inapp_banner_targets );
```

customer_id	customer_name	segment
C1001	Nomsa Zwane	Retail
C1002	Andile Buthelezi	Retail
C1003	Fatima Mahomed	SME
C1004	Ryno van Zyl	Corporate
C1005	Thandeka Cele	Retail
C1006	Samuel Nkosi	Private Wealth

Question 6 UNION ALL

The fraud team needs the complete log to all ATM transactions from machine ATM-01 and machine ATM-02. Every transaction must appear - even if the same transaction appears in both logs due to a system sync error

(5)

```
(SELECT transaction_id, account_id, amount, transaction_date
FROM atm01_transactions
UNION ALL
SELECT transaction_id, account_id, amount, transaction_date
FROM atm02_transactions );
```

transaction_id	account_id	amount	transaction_date
T1001	A001	500	2026-03-01
T1002	A002	1200	2026-03-01
T1003	A001	300	2026-03-02
T1004	A003	750	2026-03-02
T1003	A004	300	2026-03-02
T1005	A004	900	2026-03-03
T1006	A002	450	2026-03-03
T1007	A005	150	2026-03-04

Question 7

Full list of all loan application received from Gauteng and Western Cape regions. Every application must be included.

SEE CODE ON PAGE 8

application_id	customer_id	loan_type	amount_requested
LA001	C1001	Home Loan	850000
LA002	C1002	Vehicle loan	220000
LA003	C1003	Personal	45000
LA004	C1004	Home Loan	1150000
LA003	C1003	Personal	45000
LA005	C1005	Vehicle Loan	310000
LA006	C1006	Business	530000
LA007	C1007	Home Loan	920000

Question 8
List of every complaint logged this month, whether it came via email or mobile app. Every complaint must appear.

```
(SELECT complaint_id, customer_id, category, logged_date
FROM email_complaints
UNION ALL
SELECT complaint_id, customer_id, category, logged_date
FROM app_complaints);
```

Complaint_id	customer_id	category	logged_date
EC001	C2001	Card Dispute	2026-03-05
EC002	C2002	Online Banking	2026-03-06
EC003	C2003	ATM Failure	2026-03-07
EC004	C2004	Account Fees	2026-03-08
AC001	C2005	App Crash	2026-03-05
AC002	C2001	Card Dispute	2026-03-06
AC003	C2006	Payment Delay	2026-03-07
AC004	C2007	Login issue	2026-03-08

Q9 - Finance need to reconcile all payment records from April and May into one table.

```
(SELECT payment_id, account_id, amount, payment_date
FROM april_payments
UNION ALL
SELECT payment_id, account_id, amount, payment_date
FROM may_payments);
```

payment_id	account_id	amount	payment_date
PAY001	A001	12500	2026-04-15
PAY002	A002	4800	2026-04-18
PAY003	A008	9200	2026-04-25
PAY004	A004	3300	2026-04-28
PAY005	A001	12500	2026-05-15
PAY006	A005	7600	2026-05-20
PAY007	A002	5100	2026-05-22
PAY008	A006	2800	2026-05-27

Q10 The accounting system records debits and credits in two separate tables. Combine all entries into one general ledger for the audit.

```
SELECT entry_id, account_id, entry_type, amount, entry_date
FROM debit_entries
UNION ALL
SELECT entry_id, account_id, entry_type, amount, entry_date
FROM credit_entries );
```

entry_id	account_id	entry_type	amount	entry_date
DR001	A001	Debit	1500	2026-03-10
DR002	A002	Debit	3400	2026-03-11
DR003	A003	Debit	850	2026-03-12
DR004	A004	Debit	2100	2026-03-13
CR001	A001	~~Debit~~ Credit	5000	2026-03-10
CR002	A005	Credit	1200	2026-03-11
CR003	A002	Credit	4500	2026-03-12
CR004	A006	Credit	3100	2026-03-13

QUESTION 7

```
(SELECT application_id, customer_id, loan_type, amount_requested
FROM gauteng_loan_applications
UNION ALL
SELECT application_id, customer_id, loan_type, amount_requested
FROM western_cape_loan_applications );
```

SECTION 3 BONUS QUESTIONS.

1. A fraud analyst should use UNION.
Because the requirement is to get a unique list of customers who appeared in either the January or february high-risk watch lists.
⇒ UNION combines the results of two queries and remove duplicates.

2. The auditing team should use UNION ALL
Because the goal is to count every individual ~~the~~ transaction, including transactions that may appear more than once because they were recorded in multiple systems.

3. The query fails because the first SELECT returns 2 columns while the second SELECT returns 3 columns.
To fix it, I will ensure that both queries return same number of columns in the same order with compatible data types.